

LAB 0.1 · ENVIRONMENT SETUP (AUTH-TOKEN TRACK)

LocalStack with Auth Token + Web Console

A fresh, self-contained setup: activate LocalStack with a licence token, then view your local AWS resources in the browser-based Resource Browser

Course: **IKB42603 Cloud Computing Security Essentials** · Est. time: 25–30 min · Standalone (does not depend on Lab 0)

About this lab: This is the auth-token track. Since 23 March 2026 the standard LocalStack image (localstack/localstack) requires a licence auth token to start. Here you create a free account, activate LocalStack with a token, and explore your resources in the LocalStack Web Console. If you prefer a token-free setup, use the pinned Community image instead (Lab 0 + Addendum A).

Learning outcomes

By the end of this lab you will be able to: (1) obtain and safely configure a LocalStack auth token; (2) start an activated LocalStack instance in Docker; (3) confirm licence activation from the command line; (4) point the AWS CLI at LocalStack; and (5) create AWS resources and inspect them in the web-based Resource Browser.

Prerequisites

You need Docker Desktop and AWS CLI v2 installed and a modern web browser. Confirm the tools before starting:

```
docker --version      # any recent version
docker run --rm hello-world
aws --version         # expect aws-cli/2.x
```

Not installed yet? Install Docker Desktop (choose Apple Silicon or Intel to match your Mac) and AWS CLI v2 from their official sites, then re-run the checks above. Windows users: run every command in Git Bash or WSL.

Part 1 — Create an account and get an Auth Token

1. Go to <https://app.localstack.cloud/sign-up> and create a free account (student email, Google or GitHub). No credit card required.
2. Sign in, then confirm your email if prompted.
3. Open the Auth Tokens page: <https://app.localstack.cloud/workspace/auth-tokens>.
4. Copy your **Developer Token** — it looks like `1s-XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX`. This token activates the LocalStack image on your machine.

Keep it secret: An auth token grants access to your licence and workspace. Never commit it to Git, never paste it into shared documents or screenshots. If it leaks, rotate it immediately on the Auth Tokens page.

Part 2 — Set the token in your shell

Export the token as an environment variable in the terminal you will use for the lab. Replace the placeholder with your real token.

```
# macOS / Linux / Git Bash / WSL
export LOCALSTACK_AUTH_TOKEN="1s-XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX"

# Confirm it is set (should print your token)
echo $LOCALSTACK_AUTH_TOKEN
```

Tip: This variable lasts only for the current terminal session. To avoid re-typing it each time, add the export line to your ~/.zshrc (macOS) or ~/.bashrc (Linux/WSL) — but only on your personal machine, never a shared one.

Part 3 — Start LocalStack (fresh, activated)

Remove any previous LocalStack container so you start clean, then launch the activated image with your token injected.

```
# 1) Clean slate – remove any existing container (ignore "no such container")
docker rm -f localstack 2>/dev/null

# 2) Start LocalStack with the auth token
docker run -d --name localstack \
  -p 4566:4566 \
  -p 4510-4559:4510-4559 \
  -e LOCALSTACK_AUTH_TOKEN=$LOCALSTACK_AUTH_TOKEN \
  -v /var/run/docker.sock:/var/run/docker.sock \
  localstack/localstack

# 3) Watch it come up until you see "Ready."
docker logs -f localstack          # Ctrl-C to stop following
```

Why these flags: Port 4566 is the single AWS gateway; ports 4510–4559 are used by some services for dynamically opened endpoints; the docker.sock mount lets LocalStack spawn helper containers. The token is passed only as an environment variable, never baked into the image.

Part 4 — Verify licence activation

Two checks confirm a healthy, activated instance. The info endpoint proves the token worked; the health endpoint lists running services.

```
# Licence activation – edition and activation flag
curl http://localhost:4566/_localstack/info

# Expected (formatted):
#   "edition": "pro",
#   "is_license_activated": true

# Service health
curl http://localhost:4566/_localstack/health
```

Success looks like: is_license_activated is true and the container status (docker ps) shows "healthy". If instead the container Exited (55) with "License activation failed", the token is missing, wrong, or not yet assigned a licence — see Troubleshooting.

Part 5 — Point the AWS CLI at LocalStack

LocalStack accepts any credentials. Set dummy values once, then use an endpoint shortcut each session.

```
aws configure set aws_access_key_id test
aws configure set aws_secret_access_key test
aws configure set region us-east-1

# Endpoint shortcut for this session
EP='--endpoint-url=http://localhost:4566'

# Prove the CLI is talking to LocalStack
```

```
aws $EP sts get-caller-identity
```

A successful call returns a dummy identity (Account 000000000000, an example UserId, and an ARN ending in :root) — the expected result against LocalStack.

Part 6 — Open the Web Console and Resource Browser

Because your instance is activated with your token, the LocalStack Web Console recognises it as an authenticated instance and unlocks the full Resource Browser.

1. In the browser, sign in at `app.localstack.cloud` (same account as Part 1).
2. Open the **Instances** page and click the **default** instance — its endpoint `https://localhost.localstack.cloud:4566` points at your running container on this machine.
3. The Status / Stack Overview page shows the running services and confirms the console has connected to your local instance.
4. Open **Resource Browser** and browse services by category — Storage (S3), Database (DynamoDB), Security Identity & Compliance (IAM, KMS), Compute (Lambda) and more. Set the region dropdown (top-right) to `us-east-1` so your lab resources appear.

Privacy: The console page loads from the internet, but it reads your resource data directly over localhost. None of your resource data is uploaded to LocalStack servers.

Part 7 — Hands-on: create resources and see them in the console

Create a few resources from the CLI, then refresh the Resource Browser to watch them appear — the same create-then-view loop you use in the real AWS Console.

```
EP='--endpoint-url=http://localhost:4566'

# S3 bucket + a file
aws $EP s3 mb s3://ccse-demo-bucket
echo "hello cloud" > hello.txt
aws $EP s3 cp hello.txt s3://ccse-demo-bucket/

# DynamoDB table
aws $EP dynamodb create-table \
  --table-name Students \
  --attribute-definitions AttributeName=id,AttributeType=S \
  --key-schema AttributeName=id,KeyType=HASH \
  --billing-mode PAY_PER_REQUEST

# IAM user
aws $EP iam create-user --user-name lab-student
```

Now watch it live: In the Resource Browser: S3 shows `ccse-demo-bucket` (with `hello.txt` inside), DynamoDB shows the `Students` table, IAM shows the `lab-student` user. Click the Reload button after each command.

Part 8 — Stop and clean up

```
# Stop (keeps the container) / start again next session
docker stop localstack
docker start localstack

# Remove completely (clears all resources)
docker rm -f localstack
```

Removing the container is safe — recreate it any time with the Part 3 command. Your lab report files (outputs, screenshots) live on your laptop, not in the container.

Pre-lab verification checklist

Tick each item before moving on to Lab 1.

- ☐ docker --version and docker run hello-world both work.
- ☐ LOCALSTACK_AUTH_TOKEN is set (echo prints it).
- ☐ curl .../_localstack/info shows is_license_activated: true.
- ☐ curl .../_localstack/health responds with services available.
- ☐ aws \$EP sts get-caller-identity returns an identity.
- ☐ Resource Browser shows the default instance and your test resources in us-east-1.

Troubleshooting

Symptom	Fix
Container Exited (55); "License activation failed"	Token missing/invalid, or no licence assigned. Re-check echo \$LOCALSTACK_AUTH_TOKEN, re-copy from the Auth Tokens page, and confirm a licence is assigned on the My License page.
curl to :4566 fails instantly	Container is not running (it may have exited). Run "docker ps -a" and "docker logs localstack" to see why.
info shows is_license_activated: false	The token started the image but no licence is active. Assign a role/licence to your user under Workspace → Users & Licenses.
"License server unreachable" in logs	Corporate DNS may block the licence domain. Verify with "dig api.localstack.cloud"; ask your network admin to allow the localstack.cloud domain.
"Network Failure" in the console	The console cannot reach your container — start LocalStack (Part 3) and reload the page.
Console shows no resources	Wrong region — set the dropdown to us-east-1 and click Reload after creating resources.

IKB42603 Cloud Computing Security Essentials — Lab 0.1 (Auth-Token Track). Auth token required since 23 March 2026; verify activation with the /_localstack/info endpoint. Never commit tokens.